

Non-Quantum Gauge Theory in $\mathcal{UD}$

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In this notebook,

- A .wl script version of code for this notebook, minus the prose and the compiler POC, is available at

TODO asdf

Initialize the Relativity Toolkit

Optional: Quit for a Fresh Kernel

- Note, if you call **Quit** in a notebook, evaluation may hang. If you want to “Evaluate Notebook,” mark the following two cells “Evaluatable” in the Cell→Property menu, call **Quit** and **FrontEndTokenExecute** once each, then comment out the cells, or mark them unevaluatable again. Alternatively, you can quit the kernel via the Evaluation menu (last item). At that point, you can “Evaluate Notebook” in a clean environment, or evaluate cells individually.

```
In[ ]:= Quit[];
```

```
In[ ]:= FrontEndTokenExecute["DeleteGeneratedCells"]
```

Download the Code

```
In[ ]:= RTbranch = "main";
RTbase = "https://raw.githubusercontent.com/rebcabin/RelativityToolkit/" <>
RTbranch <> "/";
RTbust = "?t=" <> ToString[Round[AbsoluteTime[]]]; (*Force fresh download*)
RTrules = RTbase <> "RelativityToolkit.wl" <> RTbust;
RTtests = RTbase <> "RelativityToolkit_RegressionTests.wl" <> RTbust;
RTviz = RTbase <> "RelativityToolkit_VisualShowcase.wl" <> RTbust;
Get[RTrules];
```

» Relativity Toolkit version : 1.8.0

» Relativity Connection Symbol : Γ

Optional: See all Tests and Results from Prior Works

Optional: Configure Relativity Connection Symbol

To properly compose covariant derivatives, $\mathcal{UD}$ must know the user’s reserved symbols for non-tensorial connection coefficients. In gravitational physics, these force-producing connections are

called Γ (capital gamma). Other theories like electrodynamics and Yang-Mills that can benefit from $\mathcal{UD}$ use other connection symbols.

We're going with Γ for this relativity/gravitation notebook.

```
In[*]:= SetConnection[ $\Gamma$ ];
```

Covariant Derivatives, Revisited

So far, for general relativity (GR), we learned to take covariant derivatives of tensors that inhabit the tangent and cotangent tensor spaces of curved manifolds. The canonical example is the covariant derivative of a contravariant vector with respect to a basis of coordinate functions:

```
In[*]:= With[{example = HoldForm[CD[A[ $\mathcal{U}[\mu]$ ], x[ $\mathcal{U}[\nu]$ ]]]},  
  Echo[TensorForm@@example, example ""]];
```

$$\gg (A^{\mu}_{;\nu}) A^{\mu}_{,\nu} + (A^{\lambda})^{\mu}_{\nu} \Gamma^{\mu}_{\nu\lambda}$$

The connection coefficients $\Gamma^{\mu}_{\nu\lambda}$ account for the twisting and turning of coordinate-basis vectors in the tangent spaces as the vector A^{μ} is parallel-transported *back* from position $x^{\alpha} + dx^{\alpha}$ to position x^{α} , where we take the limit that produces the derivative.

- As mentioned in several places in the $\mathcal{UD}$ series of notebooks, the connection coefficients inform **forces**. In GR, geodesics that start off parallel converge or diverge in curved space. This **geodesic deviation** is indistinguishable from a force. See MTW.

Our covariant derivative does not depend on choice of coordinate frame: it transforms like a tensor: physics is **locally Lorentz invariant**. Quantities measured in any inertial (non-accelerating) frame $x^{\mu'}$, moving at some 3-velocity $\vec{v}$ (with $|\vec{v}| < c$, of course) transform via Lorentz or Poincaré transformation into quantities measured in another inertial frame x^{μ} .

This invariance of physics with respect to Lorentz or Poincaré transformations should remind one of Noether's theorem, by which any continuous invariance implies a conservation law. In GR, the conserved quantity is **spin current**, and arises by expressing the metric in non-coordinate bases: writing Lorentz and Poincaré invariance as local gauge invariance so as to apply Noether's theorem (see Appendix 1).

In this notebook, we consider, on top of the tensor spaces of GR, other spaces attached, like decals, to points in the curved spacetime manifold. These are spaces of gauge fields representing electromagnetism and matter. These spaces have their own differential geometries above and beyond the differential geometry of spacetime itself, which we have studied via $\mathcal{UD}$ for GR. $\mathcal{UD}$ should apply, however, to these gauge-field differential geometries just as well as $\mathcal{UD}$ applies to GR. The topic of this notebook is to apply $\mathcal{UD}$ to transformation laws, invariants, and conserved quantities of “decal” gauge fields.

- For this notebook, it is fine to envision fields of spaces “attached” to points p in the manifold, but there is a bit of jargon, **Fibre Bundles**, that one encounters in the literature. Please see Appendix 2 for the definitions and set-theoretic machinery needed to interpret this jargon.

Before leaving this introduction, let me touch upon a point of frustration. Despite many similarities, GR, as a gauge theory, is profoundly different from the others. In GR, the connection **is** the geometry,

constructed directly from the metric. Its curvature is the curvature of spacetime itself, and the forces it produces arise solely from that curvature. In electromagnetism and Yang-Mills, connections **live on top** of curved spacetime. They create stress-energy that curves the background metric, but their mathematical forms do not depend on the metric nor on the gauge fields extracted from GR. They remain distinct from the spacetime manifold, merely attached to it. This distinction—geometry as field versus field on geometry—is one of the barriers, but not the only one, keeping the holy grail of Quantum Gravity out of reach.

- You may recall from an earlier notebook that we initially derived the covariant derivative and connections without reference to a metric. It was only later, when deriving the Christoffel symbols for GR, that we invoked the metric plus constraints like torsion freedom. Relaxing torsion freedom unveils the spin connection—the gauge field of local Lorentz invariance—which is the only way GR can couple to spin density. In standard GR, this field is effectively hidden inside the Christoffel symbols. Pulling it out explicitly only strengthens the frustrating point that, unlike the **independent** gauge fields of electromagnetism and Yang-Mills, the gauge field of GR is entirely in the geometry itself.

In[] := TensorForm@CD[A[U[μ]], x[U[ν]]]

Out[] := $A^{\mu}_{\nu} + (A^{\lambda})_{\nu} \Gamma^{\mu}_{\lambda}$

In[] := CD[φ, x[U[ν]]]

Appendix 1: Spin Connection as Gravitational Gauge

This appendix is not essential to understanding the material of this notebook, but may be helpful to understanding that gravitation has a partial fit to other gauge theories of physics. “Partial” qualifies the frustration point that gravitation almost acts like any other gauge theory, but not quite. Gravitation consists *in* the geometry, while other gauge theories consist *on* the geometry.

To align gravitation with gauge theory, we are led to the conservation of spin current as a consequence of Lorentz invariance, specifically in the Einstein-Cartan-Sciama-Kibble (ECSK) theory.

Here are some references that explicitly discuss this point:

- Sean Carroll, *Spacetime and Geometry: An Introduction to General Relativity*
 - Appendix J, Page 486ff, under local Lorentz invariance, tetrads, and the spin connection: Carroll details how the spin connection ω_{μ}^{ab} , arising from non-coordinate basis sets, acts as the gauge field for local Lorentz transformations. He also explains that standard GR, via the metric formulation, assumes that the torsion tensor $T^{\lambda}_{\mu\nu}$ is zero, locking the Levi-Civita connection to the metric and forcing the energy-momentum tensor to be symmetric, absorbing spin effects via symmetrization.
 - See Hehl, the next reference, Equation 3.8 and Sections III.B and III.C. In other references, this may be called Belinfante-Rosenfeld symmetrization.
- Friedrich W. Hehl et al., *General Relativity with Spin and Torsion: Foundations and Prospects* (Rev. Mod. Phys. 48, 393, 1976)

This definitive review paper proves that:

- Invariance under general transformations (diffeomorphisms), leads to the conservation of the energy-momentum tensor.
- Invariance under local Lorentz transformations implies conservation of the spin angular momentum tensor (the spin current).
- Standard Einstein GT with vanishing torsion, which we assumed in this series of notebooks, collapses the two symmetries and hides the independent spin current.
- T. W. B. Kibble, *Lorentz Invariance and the Gravitational Field* (J. Math. Phys. 2, 212, 1961)
 - Foundational paper that derives gravity as the gauge theory of the Poincaré group, i.e., Lorentz + translations

Kibble shows that (1) gauging translations yields the tetrad field, defining the metric, coupled to energy-momentum, and that (2) gauging the Lorentz subgroup, boost + rotation, yields the spin connection, permitting transformation of spinors, otherwise impossible.

- Video: Carmeci Academy. *General Relativity | Review and Intro to The Spin Connection*, <https://www.youtube.com/watch?v=LjydxYNY7Yg>

Appendix 2: Fibre Bundles

In this appendix, we introduce some standard mathematical language, not essential for understanding the rest of this notebook, but helpful for reading other material.

Bundles

In differential geometry at-large, spaces attached to various points of the manifold are collected into sets called **bundles**. For example, if $T_p M$ (an indivisible notation) is the set of all (tangent) vectors at point p of the manifold, then the **disjoint union** TM (another indivisible notation), is defined as

$$TM \equiv \bigsqcup_{p \in M} T_p M \quad (1)$$

the set of all pairs $(p, T_p M)$ of points and tangent spaces.

Projections

Now, envision a **projection function**, π , that yields p from any element of TM :

$$\pi: TM \rightarrow M, (p, T_p M) \mapsto p \quad (2)$$

Fibres

In elementary mathematics, the **fibre**, $\text{fib}(f, p)$, of a function f at some point, p , is the preimage of f at p , $f^{-1}(p)$, that is, the subset of points of the domain of f that yield p .

- Note carefully that, in this usage, f^{-1} is not a function that yields points, but a function that yields subsets. f may not be “invertible” in the usual sense of a point-to-point bijective function, but we can always define a fibre, a set containing all the points of the domain that yield p . Unfortunately, the same notation, f^{-1} , is used both for an inverse function when f is invertible and for a fibre generally.

In the case of our projection function, the fibre is, pedantically, the pair $(p, T_p M)$, or somewhat sloppily and confusingly, just the tangent space

$$\pi^{-1}(p) \equiv T_p M \quad (3)$$

“forgetting” the point p .

Even more confusingly, some accounts “flatten” $T_p M$, defining TM as the disjoint union of all points p in the manifold and all vectors v ripped out of their tangent spaces:

$$TM \equiv \bigsqcup_{p \in M} \{v \in T_p M\} = \{(p, v) \mid p \in M, v \in T_p M\} \quad (4)$$

In which case, π^{-1} must forget both the point and some representative vector, still producing a projected subset of TM :

$$\pi^{-1}(p) = \{v \in T_p M\} \equiv T_p M \quad (5)$$

amounting to the same thing, but remembering the set of all possible vectors “attached” to point p .

Total Spaces

In this flattened case, TM is called a **total space**, usually written E .

Applied to Gauge Fields—The Principal Bundle and Sections

In the case of gravitation, the tangent spaces (in various bases) play the role of gauge fields, as discussed in Appendix 1. In the case of a physical field attached to every point of the manifold, say an electrodynamical potential $A^\mu(p)$, the gauge field is a field of symmetry groups (sets) of A^μ , say $U(1)$, one field value and one symmetry group at each point p .

If the Lagrangian density, $\mathcal{L}$, of the physics is *invariant* when symmetries from $U(1)$ are applied arbitrarily (but continuously and differentiably) to the field A^μ at each point p , $\mathcal{L}$ is **locally gauge invariant**, just as gravitation is locally Lorentz invariant. **Invariant** here means numerically equal and of the same algebraic form, as explained in my earlier notebook, *From the steam age to quantum fields*.

The collection of all symmetry spaces for all points p is the **principal bundle**, just as the collection of all tangent spaces is called the tangent bundle. The particular field, e.g., A^μ , that the gauge field modifies, is the **section** of the principal bundle.

The disjoint union of all values of the gauge fields at all points in M is the (flattened) total space, usually denoted P in this case, just as the disjoint union of all vectors at all points in M is the (flattened) total space, E , of gravitation.

$$P \equiv \bigsqcup_{p \in M} \{\varphi \in U(1)\} = \{(p, \varphi) \mid p \in M, \varphi \in U(1)\} \quad (6)$$

The fibre at each point is the preimage of a projection function, π (same symbol as used for the tangent

bundle, but only analogous meaning, here):

projection: $\pi: P \rightarrow M, (p, U(1)) \mapsto p$, forgets φ (7)

fibre: $\pi^{-1}(p) = \{\varphi \in U(1)\}$, remembers the forgotten φ (8)

References

1. Beckman, Brian. *From the steam age to quantum fields: a unified approach via UD tensorial calculus*. Wolfram Community post.
2. Beckman, Brian. *General relativity in the UD calculus*. Wolfram Community post.
3. Beckman, Brian. *Formal Differential Geometry in the UD Calculus: Part 1*. Wolfram Community post.
4. Beckman, Brian. *Covariant Derivatives and Connections in the UD Calculus*. Wolfram Community post.
5. Beckman, Brian. *Riemann Curvature in UD + Compiler POC*. Wolfram Community post (to appear).
6. Beckman, Brian. *Schwarzschild Orbits in the UD Calculus*. Wolfram Community post (to appear).
7. Beckman, Brian. *Schwarzschild Metric from Scratch via UD Compilation*. Wolfram Community post (to appear).
8. Wolfram Function Repository. *MetricTensor*. For component-based calculations.
9. Wolfram Function Repository. *ChristoffelSymbol*. For component-based calculations.
10. [MTW] Misner, Charles W.; Thorne, Kip S.; Wheeler, John Archibald. *Gravitation*. Princeton University Press, 2017.
11. Sean Carroll, *Spacetime and Geometry: An Introduction to General Relativity*. Cambridge University Press, 2019.
12. Friedrich W. Hehl et al., *General Relativity with Spin and Torsion: Foundations and Prospects* (Rev. Mod. Phys. 48, 393, 1976)
13. T. W. B. Kibble, *Lorentz Invariance and the Gravitational Field* (J. Math. Phys. 2, 212, 1961)